Reach-Avoid Reinforcement Learning with Time-Varying Safe and Target Zones

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Abstract

This work extends the **Discounted Reach-Avoid Bellman Equation (DRABE)** framework to environments with **time-varying obstacle and target sets**. By augmenting the state with time and training a **Double Deep Q-Network** over the augmented space, the learned **value function V(x,y,t)** provably tracks the instantaneous reach-avoid geometry, validated across two custom dynamic environments.

1. Introduction

Safe autonomy requires controllers that simultaneously reach a **goal T(t)** and **avoid obstacles F(t)**. Classical **Hamilton-Jacobi reachability** solves this exactly but is intractable beyond 5–6 state dimensions. **DRABE** approximates HJ via deep RL with provably conservative guarantees $\mathbf{RA_\gamma \subseteq RA}$ — but assumes static environments, failing for moving obstacles, dynamic targets, and non-stationary geometries.

This work extends DRABE to time-varying **F(t) and T(t)**. Augmenting state with time, $\tilde{s} = [x, y, t]$, transforms the non-stationary problem into a stationary one with no changes to the Bellman structure. Two custom environments with **sinusoidally oscillating obstacles** and **moving targets** are designed and evaluated.

Objectives:

- Extend DRABE to time-varying F(t), T(t) via **state augmentation**
- Design environments where **temporal reasoning** is strictly necessary
- **Validate V(x, y, t)** tracks instantaneous reach-avoid geometry
- Confirm $\mathbf{RA_\gamma \subseteq RA}$ **is preserved** under augmentation

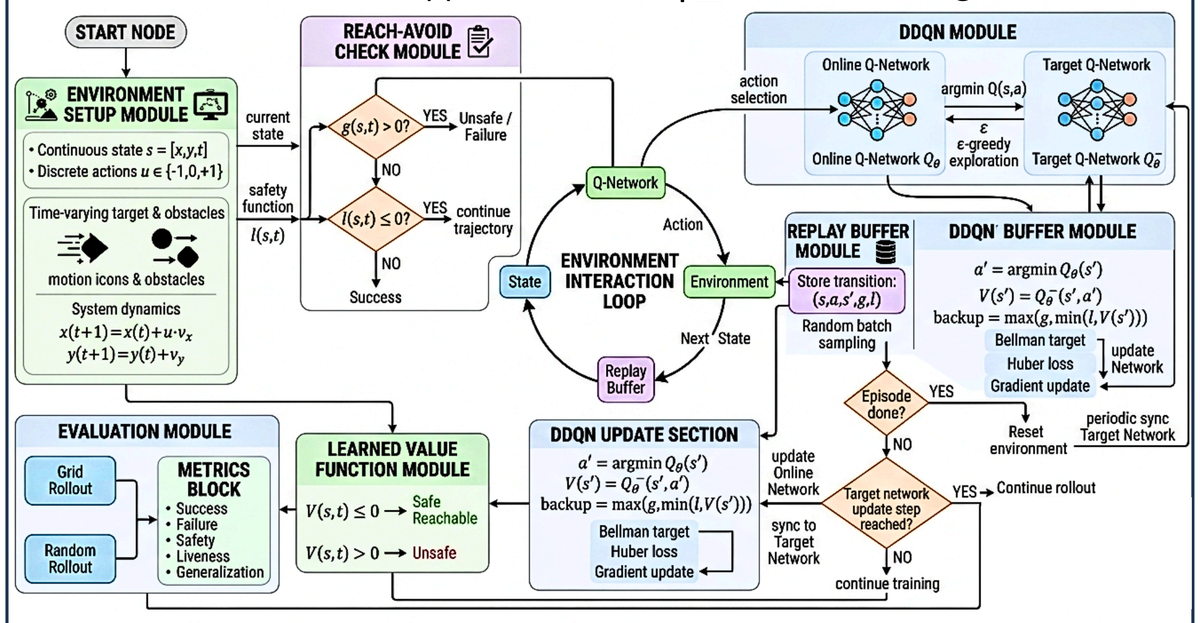
2. Methodology

The core idea is **state augmentation**: appending time to the spatial state, $\tilde{s} = [x, y, t] \in \mathbf{R^3}$, converts the non-stationary reach-avoid problem into a stationary MDP solvable by standard deep RL. Margin functions become explicitly time-dependent $\mathbf{l(\tilde{s}) \le 0 \Leftrightarrow (x,y) \in T(t)}$ and $\mathbf{g(\tilde{s}) > 0 \Leftrightarrow (x,y) \in F(t)}$ — with no modification to the Bellman operator.

Core Bellman backup (DRABE):

$$y = (1-\gamma)\max\{l(s),\ g(s)\} + \gamma \max\left\{g(s),\ \min\left\{l(s),\ Q_{\bar{w}}\left(s^u_+,\ \arg\min_{a'} Q_w(s^u_+, a')\right)\right\}\right\}$$

The **outer max enforces safety** (failure penalty), the **inner min enforces liveness** (target reward), and **γ annealed** from **0.9999→0.999999** tightens the conservative under-approximation $\mathbf{RA_\gamma \subseteq RA}$ over training.



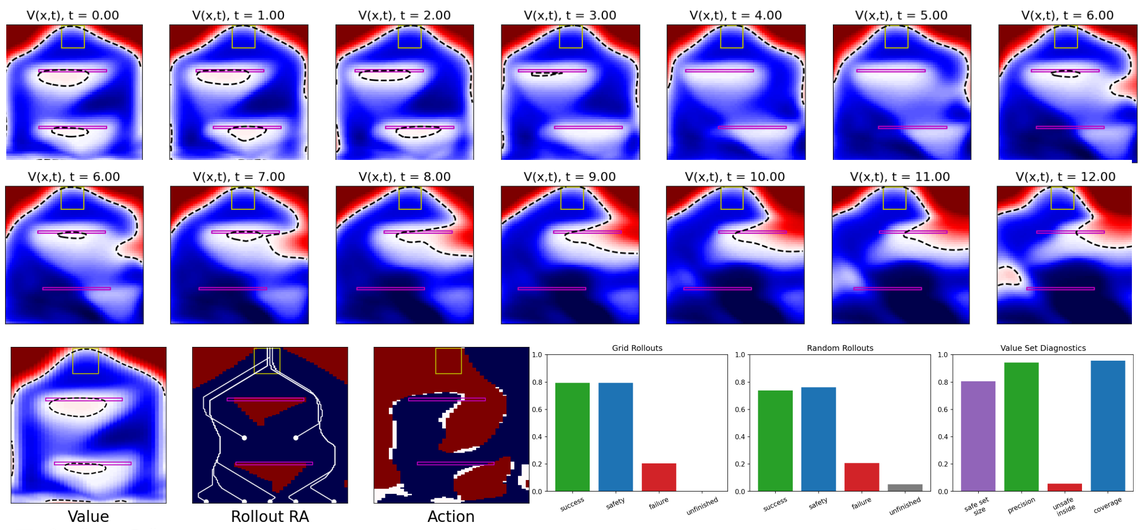
The **Q-network** maps augmented state to action values: $\mathbf{Qw : R^3 \times A \to R}$, implemented as a fully connected network with layers **[3 → 128 → 64 → 3]** and **Tanh activations**. A **warmup phase** pre-trains Qw toward max{l, g} before Bellman iterations begin.

The **online network Qw** is updated via **AdamW** with **Huber loss**, robust to outlier transitions during early exploration. The **target network** $\mathbf{Q\tilde{w}}$ is soft-updated: $\mathbf{\tilde{w} \leftarrow \tau w + (1-\tau)\tilde{w}}$ with τ = 0.01. A **replay buffer** of capacity 10,000 breaks temporal correlations by **sampling random minibatches** of size 64.

3. Results & Discussion

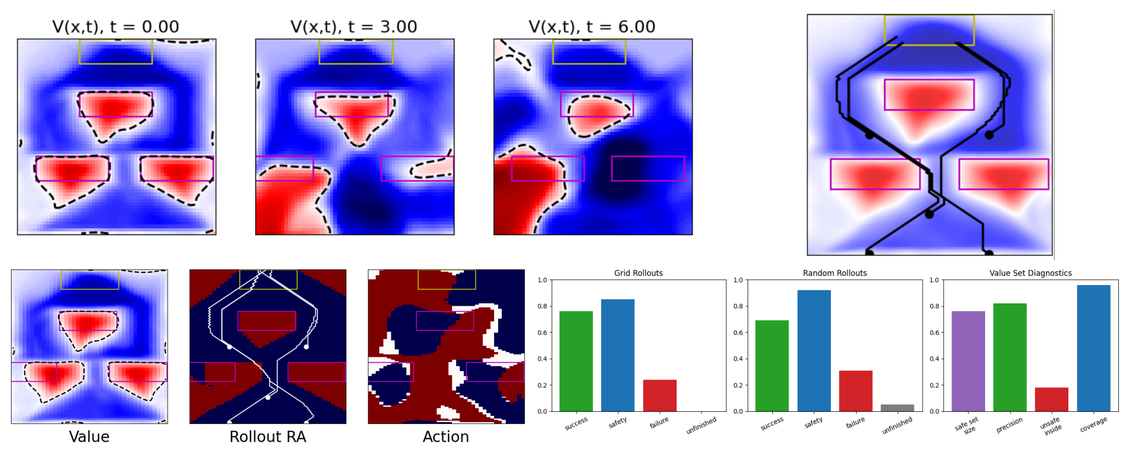
Dual-Obstacle Map:

Training converged at **4.8M updates (79.5% grid success)**. Value function time slices show the **zero sublevel set** shifting with obstacle positions, confirming time-aware learning. The near-constant level set across slices reflects the simpler obstacle geometry.



Triple-Obstacle Map:

Training converged at **6.0M updates (76.0% grid success)**. Time slices confirm the **zero sublevel set** tracks instantaneous obstacle and target positions second-by-second — validating that the agent learns a genuinely **time-conditioned policy** rather than a time-averaged one.



Quantitative comparison:

	Dual-obstacle map		Triple-obstacle map	
Metric	**Grid**	**Random**	**Grid**	**Random**
Success rate	0.795	0.739	0.760	0.690
Failure rate	0.205	0.208	0.240	0.310
Safety rate	0.795	0.762	0.760	0.690
Liveness (given safe)	1.000	0.892	1.000	0.999
Unfinished rate	0.000	0.053	0.000	0.050
Value classification (Grid only)				
Predicted RA rate	0.807		0.760	
Predicted-safe precision	0.944		0.820	
Predicted-safe FSR	0.056		0.180	
Success coverage by value	0.957		0.960	

Both maps preserve the DRABE conservativeness guarantee $\mathbf{RA_\gamma \subseteq RA}$ under time augmentation. Higher FSR on the triple-obstacle map reflects its shorter horizon and more demanding geometry, and not a failure.

4. Conclusion

State augmentation $\tilde{s} = [x, y, t]$ successfully extends DRABE to time-varying environments, converting a non-stationary reach-avoid problem into a stationary one with no changes to the Bellman structure.

- **Zero sublevel set {V(x,y,t) ≤ 0}** shifts with instantaneous obstacle and target positions, confirming time-conditioned learning
- **Conservativeness preserved: FSR as low as 5.6%**, validating $\mathbf{RA_\gamma \subseteq RA}$ under time augmentation
- Agent learns **distinct navigation strategies** at each obstacle depending on current time

References

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